Bilisuma Merga
Address: Sheger City, Oromia, Ethiopia Email: jabesabilisuma@gmail.com Phone Number: +251913182820

Education

ODA Special Boarding School, Adama, Oromia **Graduation:** August 2024
SAT Score: 1580 **Cumulative Unweighted GPA**: 95.917
Relevant Coursework: Calculus AB, Statistics, Linear Programming, Mathematical Application in Business; Applications of Physics: Two-Dimensional Motion, Fluid Mechanics, Electromagnetism, Electronics; Acid-Base Equilibria, Electrochemistry, Industrial Chemistry, Polymer, Environmental Chemistry; Application of Biology, Microorganisms, Energy Transformation, Evolution, Human Body System, Climate
Honors: Ethiopian Math Olympiad Finalist 2024; ESSLCE ranked #1 in school, #4 in Oromia, #14 in Ethiopia; China-Ethiopia Friendship Scholarship; Social Media Interview Etv News and Oromia Broadcasting Network

Skills

Technical: Statistical Modeling, Data Analysis **Scientific**: Research Conference Designing
Language: Fluent in English, Afaan Oromo, Amharic **Soft Skills:** Time Management, Leadership, Communication

Experience

Ethiopian Gifted and Talented Summer Camp| Project Developer and Trainee July 2026-September 2026

- Designed, built, and tested an **Autonomous GPS Waypoint Navigation Drone** quadcopter prototype using Arduino, GPS, IMU and IR obstacle detection.
- Developed a Python-based GIS route planner and C++ finite-state flight controller for waypoint navigation, obstacle avoidance, and Return-to-Home functionality.
- Validated the system through Proteus simulation, achieving 100% waypoint capture across 25 waypoints while using only 33% of ATmega328P SRAM.
- selected as one of the only 5 projects chosen to be showcased at the National Science Museum on **STRIDE 2.0** hosted by the **Minister of Innovation Technology**.

Teen Research Association| Curriculum Lead and Mentor October 2025 – December 2025

- Selected from 500+ applicants
- Designed Teen Research Conference 2025 pedagogy for 300+ attendees

Embao Primary School| Advisor June 2023 – August 2023

- Taught grade 7 students Math & Physics
- Improved exam Grade 8th Ministry Exam scores by 10%; awarded Top Mentor

Employment

Jeterm Trading PLC| Procurement Agent June 2024 – August 2025

- Analyzed supply costs and delivery times for 10+ contracts.
- Improved bid pricing using statistical data; generated 300k ETB monthly gains
- Analyzed tender requirements; managed logistics for specialized goods delivery

Jabesa Merga Coffee Trading| Assistant June 2021 – August 2023

- Managed 2,000kg/day distribution
- Used customer purchases to predict daily demand peaks, reducing inventory waste by 15%
- Streamlined weighing/packing workflow to increase throughput
- Generated 62.5k ETB net profit/month via demand-based pricing.

Publications & Preprints

The Cruciality of Shor's Algorithm in Breaking RSA Encryption SSRN, November 2024

Link: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4992090

- Posted review paper on SSRN and ranked Top 10 theoretical CS downloads in first month
- Independently studied RSA cryptography and Shor's quantum period-finding algorithm
- Authored a structured explanatory paper synthesizing sources from *arXiv* and graduate-level lecture notes
- Analyzed how quantum algorithms undermine classical public-key encryption and motivate post-quantum cryptography
- As of December 2024, viewed by **1,357 scholars** and downloaded by **155 scholars**

Certificates

STEM Foka Center| Participant January 2022 – June 2022

- Took course on Electronics & Programming and learned advanced circuits & its physics
- Passed the final test; received an award for summer camp

Interests: Quantum Computing, Applied Physics, Educational Equity, Playing Football, Star Wars Movies, Coffee Culture

Personal portfolio link: https://bilimarga.github.io/portfolio/